

Introduction & Purpose

The purpose of this project is to develop a failed-login monitoring and analysis tool that enhances visibility into authentication-related security events. In many traditional systems, failed login attempts are either minimally logged or ignored entirely, which limits their usefulness for both security monitoring and digital forensic investigations. This project addresses that gap by capturing detailed information about login attempts and transforming it into actionable intelligence for administrators.

In the context of digital forensics, authentication logs are often one of the earliest indicators of malicious activity. Attackers commonly rely on techniques such as brute-force attacks, credential stuffing, and account enumeration, all of which generate repeated login failures. Without structured logging and analysis, these behaviors can go unnoticed. This tool solves that problem by systematically recording login attempts, tracking patterns of failure, and identifying suspicious behavior based on timing, frequency, and password characteristics.

This project contributes to the field of cybersecurity by demonstrating how authentication systems can serve as both access control mechanisms and security sensors. Instead of simply rejecting incorrect passwords, the system captures metadata such as IP addresses, timestamps, and attempt counts, then analyzes that data to produce risk-based alerts. As a result, the tool supports early attack detection, improves incident response, and provides valuable forensic evidence for later investigation.

Technical Implementation

The system is built using Python and follows a modular architecture that separates core security logic from user interfaces. At the center of the system is a shared engine (`security_tool.py`), which handles authentication, logging, analysis, and account lockout logic. This core module is used by both a command-line interface (`setup.py`) and a web-based interface implemented with Flask (`app.py`). The frontend consists of simple HTML pages, including a login page and an administrative dashboard.

The web version of the application captures IP addresses automatically from incoming HTTP requests, using either the `X-Forwarded-For` header or the remote address of the request. This allows the system to associate login attempts with specific sources, which is critical for identifying suspicious activity. All login events are stored in JSON format (`login_logs.json`), while account state information such as failed-attempt counts and lockout

status is stored separately in `auth_state.json`. A human-readable report is generated in a text file for administrative review.

Several important techniques are implemented within the system. Passwords are securely stored using hashing, ensuring that plaintext credentials are not exposed. Each login attempt is logged with detailed metadata, including timestamps, usernames, IP addresses, success or failure status, and password similarity scores. The password similarity feature uses edit-distance logic to measure how closely a failed password resembles the correct one, providing additional context about whether an attack may be random or targeted.

The system also includes behavioral analysis capabilities. It evaluates how frequently login attempts occur, how many users are targeted by a single IP address, and how quickly repeated attempts are made. These factors are combined into a rule-based risk scoring system that classifies activity into severity levels such as LOW, MEDIUM, HIGH, or CRITICAL. A key design decision was to use a rule-based model instead of machine learning, as it provides transparency and predictable behavior while still delivering meaningful insights.

Another important feature is the account lockout mechanism. After a configurable number of failed attempts (set to five in this project), the system automatically locks the account to prevent further unauthorized access. An administrative unlock feature allows recovery while maintaining security control. Overall, the architecture emphasizes simplicity, modularity, and clear separation of responsibilities, making the system easy to extend in the future.

Results & Analysis

The tool produces structured logs and summarized reports that provide clear insight into authentication activity. Each login attempt is recorded as a JSON object containing relevant forensic details such as the time of the attempt, the username involved, the source IP address, and whether the attempt was successful. Additional fields, such as the number of failed attempts and password similarity percentage, provide deeper context for analysis.

From these logs, the system generates administrator-facing alerts that summarize suspicious activity in a readable format. For example, an alert might indicate that a specific IP address generated a high number of failed attempts in a short period of time, targeted multiple user accounts, and used passwords with high similarity to the correct credentials. Based on this information, the system classifies the activity as high risk and recommends actions such as blocking the IP address or resetting affected accounts.

The results demonstrate that the tool is effective at identifying common attack patterns. Rapid login attempts with very short time intervals often indicate automated scripts or bots, while repeated attempts against multiple users suggest credential stuffing. High similarity scores between attempted and actual passwords may indicate targeted attacks or prior knowledge of user credentials. By combining these signals, the system is able to prioritize threats and reduce the burden on administrators, who would otherwise need to manually review large volumes of raw log data.

Lessons Learned & Conclusion

Throughout the development of this project, several important lessons were learned. One of the most significant successes was the use of a modular architecture, which made it easier to develop, test, and extend different parts of the system independently. The decision to log data in JSON format also proved useful, as it allowed for easy inspection and debugging while maintaining a structured format suitable for analysis. Additionally, the inclusion of password similarity scoring added a layer of insight that goes beyond simple success or failure, helping to distinguish between random errors and more sophisticated attack attempts.

However, the project also presented several challenges. Designing a risk scoring system that was both simple and meaningful required careful consideration, as overly complex models can become difficult to interpret, while overly simple ones may miss important patterns. Another challenge was balancing realism with the constraints of a prototype system. For example, the password similarity feature relies on knowledge of the correct password, which would not be appropriate in a real-world production environment. Handling IP address extraction reliably across different deployment scenarios also required attention.

There are several aspects of the system that could be improved. The use of hardcoded demo credentials limits realism, and the administrative dashboard currently lacks proper authentication controls. Additionally, storing logs in JSON files is not scalable for large systems, where a database or centralized logging platform would be more appropriate. The absence of real-time alerting mechanisms, such as email or SMS notifications, also limits the system's responsiveness.

If the project were to be developed further, several enhancements would be implemented. These include migrating to a database-backed storage system, adding role-based access control for administrators, and integrating with security information and event management (SIEM) platforms. Additional features such as IP blocking, geolocation analysis, and multi-factor authentication would further strengthen the system. Incorporating machine learning techniques using libraries like scikit-learn could also enable more advanced anomaly detection.

In conclusion, this project demonstrates how a simple authentication system can be expanded into a powerful security monitoring tool. By capturing and analyzing failed login attempts, the system provides both defensive capabilities, such as account lockouts, and investigative value through detailed logs and risk-based alerts. While the current implementation is a prototype, it establishes a strong foundation for future development and highlights the importance of treating authentication as a key source of security intelligence rather than just an access control mechanism.

Appendix

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Failed Login Monitoring Report
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Total Events: 24
Failed Attempts: 16
Locked Attempts: 1
Successful Logins: 3
Unknown User Attempts: 1

Suspicious IP Analysis
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IP Address: 127.0.0.1
Severity: CRITICAL (score 105)
Attempts: 11
Users Targeted: Jonah, lddunke
Average Seconds Between Attempts: 97.87
Closest Password Similarity: 16.67%
Admin Message: [CRITICAL] IP 127.0.0.1 generated 11 failed attempts against Jonah, lddunke; average gap 97.87 seconds; closest attempted password matched 16.67% of the real password pattern. Recommended action: review the source IP, keep impacted accounts locked if needed, and reset credentials when similarity is unusually high. Risk score: 105.

IP Address: 192.168.1.50
Severity: CRITICAL (score 85)
Attempts: 6
Users Targeted: user1
Average Seconds Between Attempts: 0.04
Closest Password Similarity: 40.0%
Admin Message: [CRITICAL] IP 192.168.1.50 generated 6 failed attempts against user1; average gap 0.04 seconds; closest attempted password matched 40.0% of the real password pattern. Recommended action: review the source IP, keep impacted accounts locked if needed, and reset credentials when similarity is unusually high. Risk score: 85.

IP Address: 10.0.0.31
Severity: LOW (score 22)
Attempts: 1
Users Targeted: Laney
Average Seconds Between Attempts: None
Closest Password Similarity: 75.0%
Admin Message: [LOW] IP 10.0.0.31 generated 1 failed attempts against Laney; single attempt observed; closest attempted password matched 75.0% of the real password pattern. Recommended action: review the source IP, keep impacted accounts locked if needed, and reset credentials when similarity is unusually high. Risk score: 22.

Detailed Log Entries
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Logs for the failed attempts.

Admin Dashboard

Review locked accounts, generate reports, and inspect suspicious IP behavior.

[Back to login page](#)

Locked Accounts

Jonah

Locked at: 2026-04-14T23:23:27.237940

Reason: 5 failed attempts

Unlock User

Daniela ▾

Unlock Account

Report Actions

Generate the text report file used for administration analysis.

Generate Logs.txt

Current Totals

Total events: 23

Failed attempts: 16

Locked attempts: 1

Successful logins: 3

Admin dashboard with a locked user account and multiple events in the report.

Failed Login Monitoring Tool

This web server records the visitor IP automatically during login attempts, applies account lockouts after 5 failed tries, and surfaces suspicious patterns for admins.

User Login

Use one of the demo users below or type another username to simulate an unknown account attempt.

Username

user1

Password

Enter password

Attempt Login

[Open admin dashboard](#)

Live Monitoring Snapshot

Total Events

18

Failed Attempts

11

Locked Attempts

1

Unknown Users

1

Suspicious IP Alerts

CRITICAL from 192.168.1.50

[CRITICAL] IP 192.168.1.50 generated 6 failed attempts against user1; average gap 0.04 seconds; closest attempted password matched 40.0% of the real password pattern. Recommended action: review the source IP, keep impacted accounts locked if needed, and reset credentials when similarity is unusually high. Risk score: 85

This is the monitoring tool with the user login and the snapshot of the live report.